



# Ransomware Simulation and Verified Recovery

## Project 4 - Unified Evidence Portfolio

**Prepared by:** Anyaugo Space-Tech Ltd.

**Environment:** Ubuntu and Kali in a controlled VirtualBox lab

**Classification:** Public portfolio sample / capability demonstration

**Evidence:** 15 images across the numbered project sections

**Date:** 16 September 2026

### Business question

If important files become unavailable, can we preserve the evidence, restore a checked working copy and explain the result clearly?

### Verified result

Five synthetic invoices were renamed, then restored from backup. The archive checksum passed, all twenty invoices matched the backup manifest, and the five marked copies remained preserved.

### Portfolio boundary

This was a harmless filename-change exercise, not real ransomware or a client incident. The report distinguishes observed lab results from the proposed workflow for an authorized client engagement.

# Evidence Index

Each test begins with its purpose, action, result and relevance to client work. Evidence follows on separate pages so the original screenshots remain readable.

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## Start with the result

For a short business review, read the cover and Test 10. Use Tests 01-09 to inspect the supporting evidence and Test 11 for the proposed privacy and AI-use approach.

Prepared records and recommendations are labelled separately from original lab screenshots. The PDF preserves the source images without retouching their contents.

# Scope and Authorization

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## Control objective

Define exactly which systems, files and activities belong to the exercise before any simulation starts.

## What was done

The scope record names the Ubuntu victim, Kali testing machine, internal lab network and synthetic business files. It excludes client systems, public targets, real personal data and destructive malware.

## What the evidence shows

A written lab boundary is documented. This record is a training-scope statement; it is not a signed client authorization letter and does not establish permission over any third-party asset.

## How this applies to client work

For client work, obtain written permission from the asset owner or an authorized representative. Agree targets, methods, access, timing, exclusions, incident contacts and stop conditions. Provider terms and any third-party dependencies also need checking. See reference R1.

## What this evidence does not establish

Industry guidance describes good testing practice. It does not grant blanket legal authorization. Company branding identifies the report's author, not an external approving authority.

**Supporting evidence: 1 image on the following page.**

## Scope and Authorization

### Lab scope and authorization record

The prepared record defines the training boundary and the exclusions.

## Project 4 Scope and Authorization Record

Controlled ransomware simulation and recovery lab

Test 01 Neutral lab evidence Before simulation

### Business purpose

This record defines the agreed boundary for a controlled Project 4 training lab. The exercise demonstrates how important business files can be monitored, disrupted in a harmless simulation, contained, restored, and reported without touching real client systems or public targets.

#### Authorization basis

Testing remains inside a controlled training lab and follows the permitted lab boundaries.

#### Systems in scope

Ubuntu victim, Kali testing machine, internal Project 4 network, and synthetic business files.

#### Main outcome

Show preparation, detection, containment, recovery, and plain-language reporting.

### Approved activities

- Prepare and verify recovery backups before the simulation.
- Record file activity on the Ubuntu victim system.
- Run a harmless controlled disruption against synthetic lab files only.
- Investigate what changed, contain the activity, restore files, and confirm recovery.
- Explain results in a clear business-facing report.

### Excluded activities

- No public websites, client systems, production systems, or unrelated home files.
- No real personal data, financial records, customer data, or confidential company data.
- No real ransomware, destructive malware, persistence, credential theft, or evasion.
- No work outside the agreed VirtualBox Project 4 lab boundary.

### Professional authorization note

This record does not claim client or company authorization. It documents that the exercise stays inside a controlled training lab and follows the permission boundaries of the lab environment. In a real client engagement, written permission must come from the organization or authorized asset owner before testing, monitoring, incident-response activity, or recovery work begins.

#### Industry reference

This page follows the idea of Rules of Engagement in NIST SP 800-115: define the guidelines, limits, and authority for security testing before the work starts. It is a scope record for a training lab, not a legal opinion.

Evidence label: 1A-Scope-And-Authorization-Record

Status: lab scope defined before the controlled simulation evidence is collected

Project 4 controlled lab evidence

# Lab Setup and Backups

---

## Control objective

Prepare a controlled environment and a checked recovery source before altering the synthetic files.

## What was done

Two VirtualBox machines were configured on the internal P04-IR-Lab network. Ubuntu holds the sample invoices; Kali provides the testing workstation. A baseline snapshot and a compressed archive of twenty invoices were prepared.

## What the evidence shows

The evidence shows both machines configured and running, an Ubuntu baseline snapshot present, and backup verification output. The final Test 09 recovery provides the strongest confirmation: the archive checksum and all twenty restored-set file checks passed.

## How this applies to client work

Agree which business data and services must be recovered first. Test a restoration before an exercise. Maintain recovery copies with separation appropriate to the client's risks, such as offline or immutable copies with restricted administration. See reference R3.

## What this evidence does not establish

A VM snapshot is a rollback checkpoint, not an independent backup. The preserved Windows archive is outside the VM but on the same physical PC; it does not demonstrate protection against loss of that PC. Snapshot restoration was not exercised here.

**Supporting evidence: 6 images on the following pages.**

## Lab Setup and Backups

### 2A - Ubuntu lab configuration

Ubuntu is assigned 4 GB memory, two processors, a 40 GB virtual disk and the internal lab network.

**Details**



**General**

Name: Project04-Ubuntu-Victim  
Operating System: Ubuntu (64-bit)

**System**

Base Memory: 4096 MB  
Processors: 2  
Boot Order: Optical, Hard Disk  
Acceleration: Nested Paging, KVM  
Paravirtualization

**Preview**



**Storage**

Controller: IDE  
IDE Primary Device 0: [Optical Drive] VBoxGuestAdditions.iso (50.92 MB)  
Controller: SATA  
SATA Port 0: Project04-Ubuntu-Victim.vdi (Normal, 40.00 GB)

**Audio**

Host Driver: Default  
Controller: ICH AC97

**Network**

Adapter 1: Intel PRO/1000 MT Desktop (Internal Network, 'P04-IR-Lab')

**USB**

USB Controller: OHCI, EHCI  
Device Filters: 0 (0 active)

**Shared folders**

None

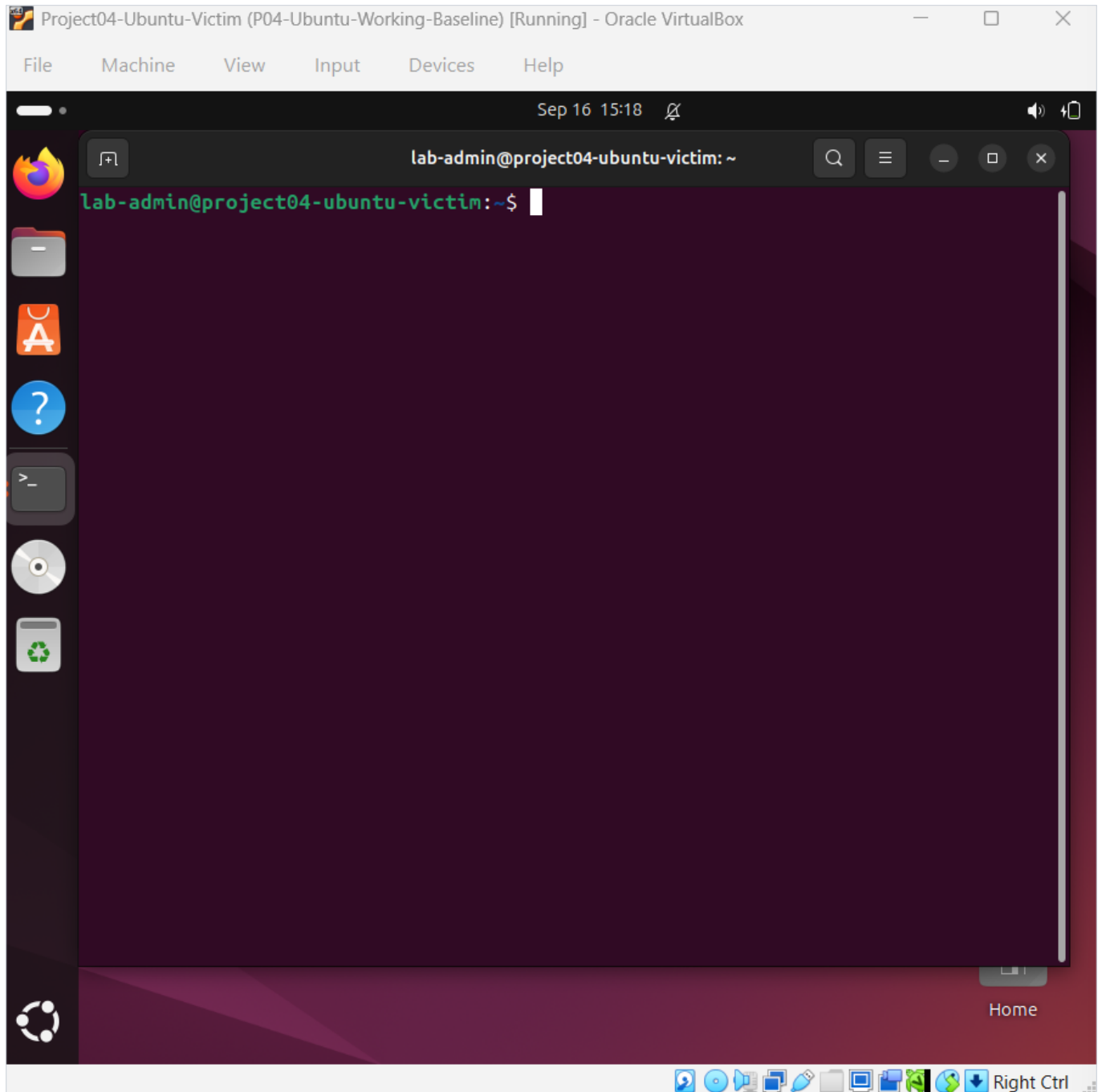
**Description**

None

## Lab Setup and Backups

### 2B - Ubuntu running

The Ubuntu VM and the lab-admin terminal session are running. Service readiness is verified separately in Test 03.



## Lab Setup and Backups

### 2C - Kali lab configuration

Kali is assigned to the same internal network. Its second adapter is shown as not attached.

**Details**



**General**

Name: Project04-Kali-Attacker  
Operating System: Debian (64-bit)

**System**

Base Memory: 4096 MB  
Processors: 2  
Boot Order: Floppy, Optical, Hard Disk  
Acceleration: Nested Paging, KVM  
Paravirtualization

**Preview**



**Storage**

Controller: IDE  
IDE Primary Device 0: [Optical Drive] Empty  
Controller: SATA  
SATA Port 0: Project04-Kali-Attacker.vdi (Normal, 40.00 GB)

**Audio**

Host Driver: Default  
Controller: ICH AC97

**Network**

Adapter 1: Intel PRO/1000 MT Desktop (Internal Network, 'P04-IR-Lab')  
Adapter 2: Intel PRO/1000 MT Desktop (Not attached)

**USB**

USB Controller: OHCI, EHCI  
Device Filters: 0 (0 active)

**Shared folders**

None

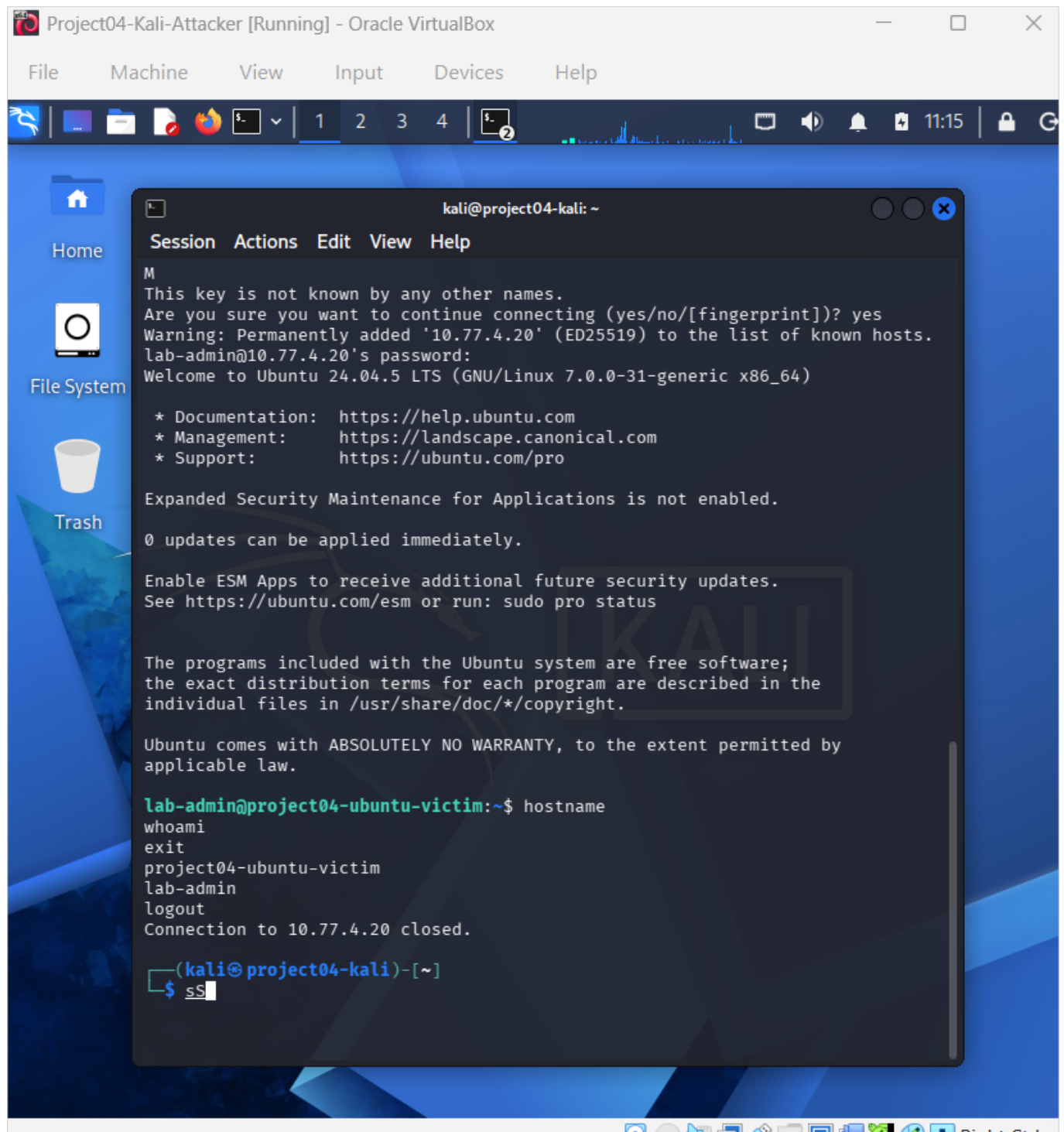
**Description**

None

## Lab Setup and Backups

### 2D - Kali running and reaching Ubuntu

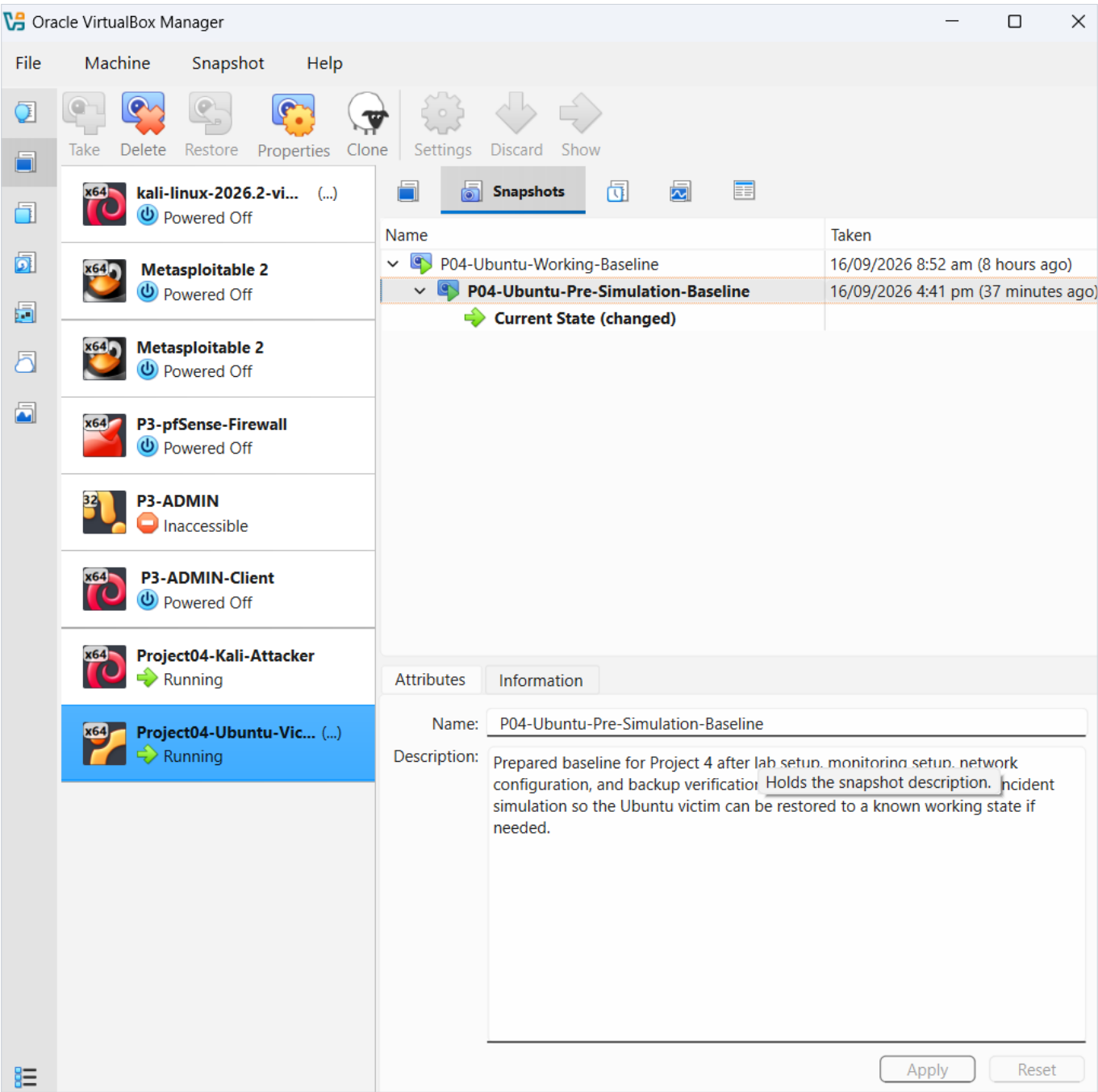
An SSH session from Kali reaches the Ubuntu lab account and then exits. This demonstrates access with supplied credentials, not exploitation.



# Lab Setup and Backups

## 2E - Ubuntu baseline snapshot

VirtualBox lists the pre-simulation baseline. This screenshot records the restore point's presence, not a completed snapshot restore.



2E-Ubuntu-Pre-Simulation-Baseline-Snapshot.png

## Lab Setup and Backups

### 2F - Pre-exercise backup check

The visible portion shows invoice checks passing and an archive checksum. Test 09 later displays verification of all twenty invoices after recovery.

---

```
invoice-06.txt: OK
invoice-07.txt: OK
invoice-08.txt: OK
invoice-09.txt: OK
invoice-10.txt: OK
invoice-11.txt: OK
invoice-12.txt: OK
invoice-13.txt: OK
invoice-14.txt: OK
invoice-15.txt: OK
invoice-16.txt: OK
invoice-17.txt: OK
invoice-18.txt: OK
invoice-19.txt: OK
invoice-20.txt: OK

Backup: /var/backups/project04/business-files-20260916T082727Z.tar.gz
Restore test: /var/tmp/p04-restore-test.Yy32wP
lab-admin@project04-ubuntu-victim:~$ (
set -eu
name="business-files-20260916T082727Z.tar.gz"
sudo install -m 600 -o "$(id -u)" -g "$(id -g)" \
  "/var/backups/project04/$name" "$HOME/$name"
sha256sum "$HOME/$name"
)
70bced19594c058997f7d340971a6efe73083ce3695f873874bd9a7a5a8176d5  /home/lab-admin/business-files-20260916T082727Z.tar.gz
lab-admin@project04-ubuntu-victim:~$ S
```

# Ubuntu Endpoint Setup

---

## Control objective

Confirm that the machine holding the files can support the planned monitoring and controlled remote access.

## What was done

Ubuntu identity and the lab account were checked. The SSH service and audit service were queried, and the loaded file-monitoring rules were listed.

## What the evidence shows

The screenshot identifies project04-ubuntu-victim and lab-admin. Both SSH and auditd report active. Audit rules cover the lab data directory and use the p04\_files key to identify relevant records.

## How this applies to client work

An analyst would verify the approved endpoint, access method, logging coverage, time settings and evidence retention before starting. The system owner would approve installation or configuration changes through the client's normal change process.

## What this evidence does not establish

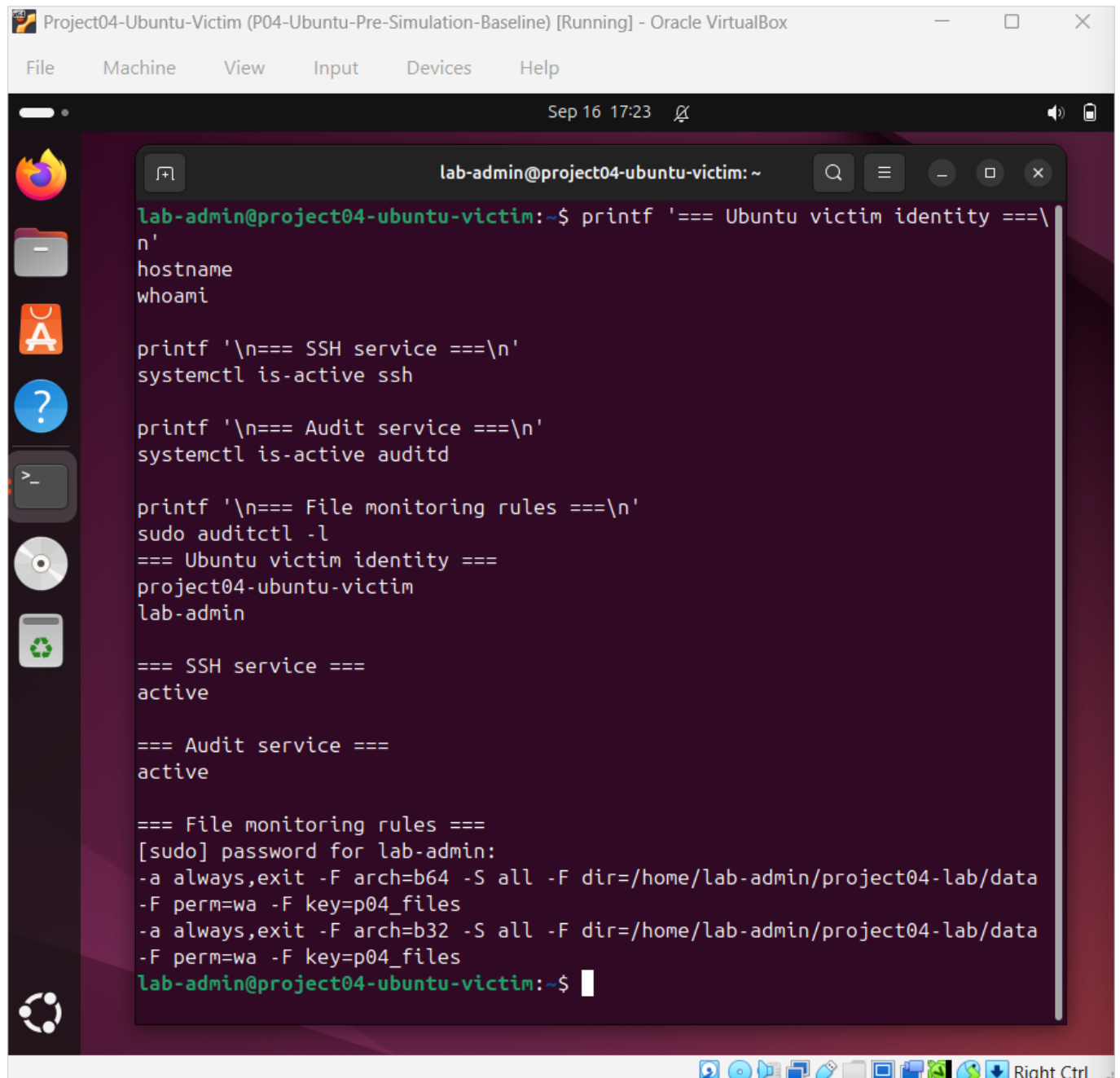
An active service and loaded rules show readiness. They do not establish complete monitoring coverage, automated alerting or protection against all file changes. Test 04 separately validates one recorded event.

**Supporting evidence:** 1 image on the following page.

## Ubuntu Endpoint Setup

### Ubuntu endpoint readiness

Identity, active services and loaded audit rules are visible together.



```
Project04-Ubuntu-Victim (P04-Ubuntu-Pre-Simulation-Baseline) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 16 17:23

lab-admin@project04-ubuntu-victim: ~
lab-admin@project04-ubuntu-victim:~$ printf '=== Ubuntu victim identity ===\n'
hostname
whoami

printf '\n=== SSH service ===\n'
systemctl is-active ssh

printf '\n=== Audit service ===\n'
systemctl is-active auditd

printf '\n=== File monitoring rules ===\n'
sudo auditctl -l
=== Ubuntu victim identity ===
project04-ubuntu-victim
lab-admin

=== SSH service ===
active

=== Audit service ===
active

=== File monitoring rules ===
[sudo] password for lab-admin:
-a always,exit -F arch=b64 -S all -F dir=/home/lab-admin/project04-lab/data
-F perm=wa -F key=p04_files
-a always,exit -F arch=b32 -S all -F dir=/home/lab-admin/project04-lab/data
-F perm=wa -F key=p04_files
lab-admin@project04-ubuntu-victim:~$
```

# File Monitoring and Detection

---

## Control objective

Check that an expected change to a monitored file produces evidence that can be reviewed.

## What was done

A harmless monitoring-check file was created in the lab data directory. The audit records for that file were retrieved and inspected.

## What the evidence shows

The record contains the file path, a successful file-open/create operation, the bash process, lab user context and the p04\_files monitoring key. This proves that the controlled test event was recorded.

## How this applies to client work

Use an agreed test event to validate collection end to end: endpoint record, central ingestion where applicable, detection rule, analyst notification and response ownership. Check normal business activity to reduce unnecessary alerts.

## What this evidence does not establish

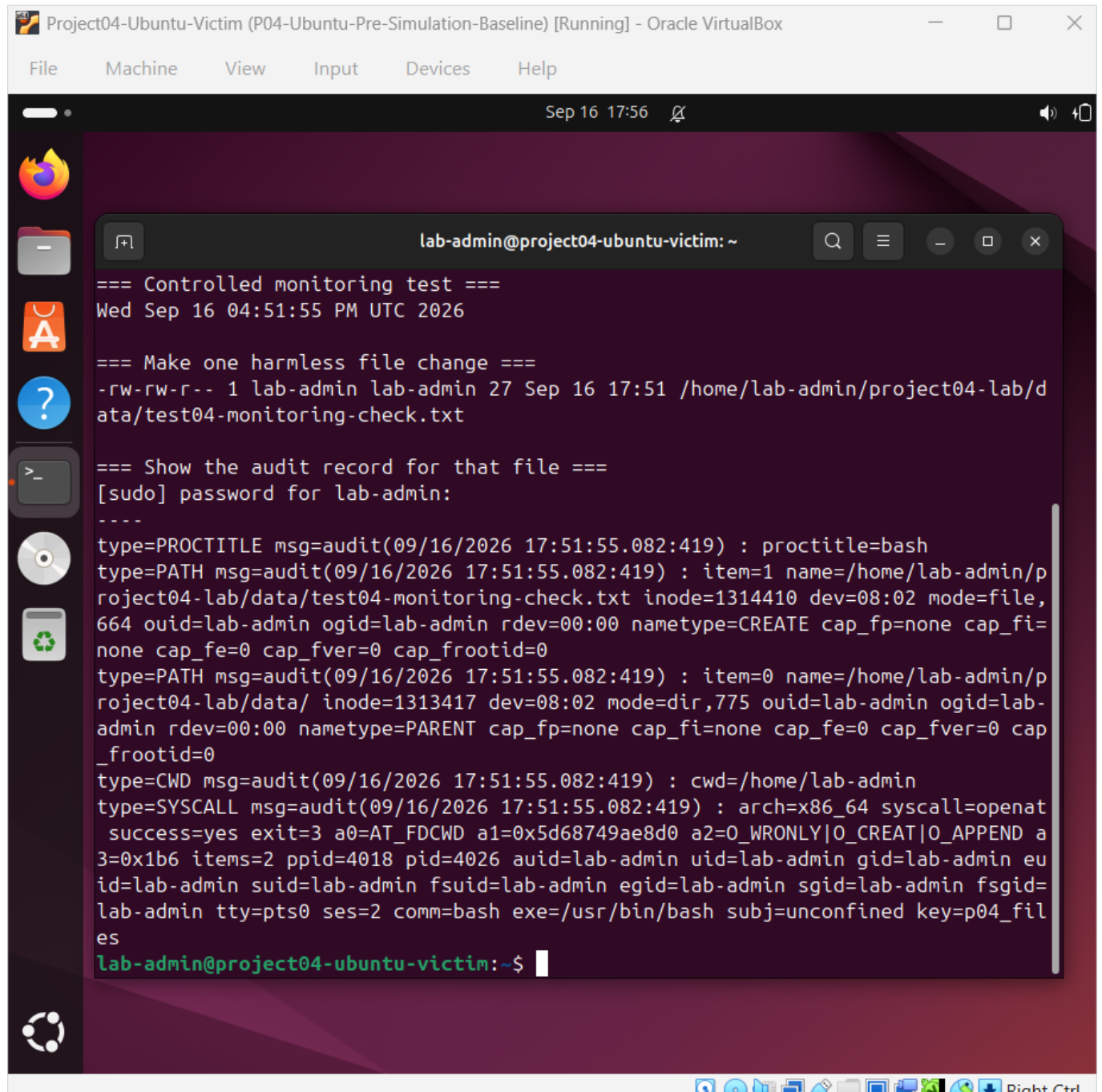
This is file-event logging evidence. It does not by itself demonstrate a ransomware detection rule or an alert for the five invoice renames performed later in Test 07.

**Supporting evidence: 1 image on the following page.**

## File Monitoring and Detection

### Controlled file change and audit record

A successful monitored file operation is tied to the lab user and the expected audit key.



```
Project04-Ubuntu-Victim (P04-Ubuntu-Pre-Simulation-Baseline) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 16 17:56

lab-admin@project04-ubuntu-victim: ~
=== Controlled monitoring test ===
Wed Sep 16 04:51:55 PM UTC 2026

=== Make one harmless file change ===
-rw-rw-r-- 1 lab-admin lab-admin 27 Sep 16 17:51 /home/lab-admin/project04-lab/data/test04-monitoring-check.txt

=== Show the audit record for that file ===
[sudo] password for lab-admin:
----
type=PROCTITLE msg=audit(09/16/2026 17:51:55.082:419) : proctitle=bash
type=PATH msg=audit(09/16/2026 17:51:55.082:419) : item=1 name=/home/lab-admin/project04-lab/data/test04-monitoring-check.txt inode=1314410 dev=08:02 mode=file,664 ouid=lab-admin ogid=lab-admin rdev=00:00 nametype=CREATE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(09/16/2026 17:51:55.082:419) : item=0 name=/home/lab-admin/project04-lab/data/ inode=1313417 dev=08:02 mode=dir,775 ouid=lab-admin ogid=lab-admin rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(09/16/2026 17:51:55.082:419) : cwd=/home/lab-admin
type=SYSCALL msg=audit(09/16/2026 17:51:55.082:419) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x5d68749ae8d0 a2=O_WRONLY|O_CREAT|O_APPEND a3=0x1b6 items=2 ppid=4018 pid=4026 auid=lab-admin uid=lab-admin gid=lab-admin euid=lab-admin suid=lab-admin fsuid=lab-admin egid=lab-admin sgid=lab-admin fsgid=lab-admin tty=pts0 ses=2 comm=bash exe=/usr/bin/bash subj=unconfined key=p04_files
lab-admin@project04-ubuntu-victim:~$
```

# Alert Review and Reporting

---

## Control objective

Translate the technical monitoring result into a finding a business reader can understand.

## What was done

The Test 04 event was reviewed and written up with its context, significance and next response step.

## What the evidence shows

The activity was classified as expected lab activity. The prepared page explains that file activity is visible for investigation without labelling the harmless test as an attack.

## How this applies to client work

An analyst would validate the signal against user activity, change approvals and other evidence. A useful finding identifies affected assets, likely impact, confidence, an accountable owner and the next action. Incident decisions should follow the agreed response process. See reference R2.

## What this evidence does not establish

The page is an analyst-style interpretation of a known test event. It is not evidence that an automated alert was delivered or that the later simulation was detected automatically.

**Supporting evidence: 1 image on the following page.**

## Alert Review and Reporting

### Monitoring review and business explanation

The prepared finding explains a known benign event and its evidential value.

## Project 4 Alert Review and Reporting

Turning raw monitoring evidence into a business-readable finding

Test 05

Review and report

Monitoring validated

### Finding summary

A controlled file change was recorded inside the monitored Project 4 data folder. The event was expected because it was created for the lab, but it proves the monitoring control can record a file action that an analyst can review and explain.

#### What happened

A harmless test file was created or updated inside the monitored Ubuntu data folder.

#### Why it matters

The audit trail shows the file path, action, user context, process, and monitoring key.

#### Analyst judgement

Expected lab activity. No real incident is claimed from this evidence alone.

### Evidence interpreted

- The monitored folder generated an audit record when the test file changed.
- The record included the Project 4 audit key, linking the event to the file-monitoring rule.
- The activity was performed by the lab user during a controlled test, so it is treated as expected activity.
- In a live client environment, a similar alert would be reviewed for business impact, affected files, user context, and containment needs.

### Business-readable report wording

Monitoring confirmed that file activity inside the protected lab folder is visible for investigation. The test did not prove an attack; it proved that the endpoint is producing useful evidence before the simulation begins.

### Next response step

Move to network isolation evidence, then run the harmless disruption simulation only inside the agreed Project 4 lab boundary.

Evidence label: Test-05-Alert-Review-And-Reporting  
Status: Test 04 monitoring evidence reviewed and explained

Project 4 controlled lab evidence

# Network Isolation Evidence

---

## Control objective

Check lab connectivity and the absence of an ordinary default route from the testing workstation.

## What was done

Kali's identity, interface addresses and routes were inspected. Three test packets were sent to Ubuntu inside the lab network.

## What the evidence shows

Kali uses 10.77.4.10/24 and reaches Ubuntu at 10.77.4.20 with three replies and no packet loss. Its second interface is down. The output reports no default IPv4 or IPv6 route.

## How this applies to client work

Before a simulation, validate segmentation and permitted paths using the client's approved network controls. During a real incident, any endpoint isolation should be coordinated with the incident lead and service owner, then checked to confirm the intended boundary.

## What this evidence does not establish

Together with the VirtualBox settings, this supports the configured lab boundary at capture time. It is not proof of an air gap, a complete connectivity audit or incident-time containment of a compromised host.

**Supporting evidence: 1 image on the following page.**

## Network Isolation Evidence

### Kali network checks

The two lab machines communicate while Kali shows no default route in the displayed routing checks.

```
kali@project04-kali: ~  
Session Actions Edit View Help  
hostname  
whoami  
  
printf '\n=== Kali network address ===\n'  
ip -br address  
ip route  
ip -6 route  
  
printf '\n=== Reach Ubuntu victim inside lab ===\n'  
ping -c 3 10.77.4.20  
  
printf '\n=== No default internet route expected ===\n'  
ip route | grep -E '^default' || echo 'No IPv4 default route'  
ip -6 route | grep -E '^default' || echo 'No IPv6 default route'  
=== Kali lab network identity ===  
project04-kali  
kali  
  
=== Kali network address ===  
lo UNKNOWN 127.0.0.1/8 ::1/128  
eth0 UP 10.77.4.10/24  
eth1 DOWN  
10.77.4.0/24 dev eth0 proto kernel scope link src 10.77.4.10 metric 100  
  
=== Reach Ubuntu victim inside lab ===  
PING 10.77.4.20 (10.77.4.20) 56(84) bytes of data.  
64 bytes from 10.77.4.20: icmp_seq=1 ttl=64 time=10.1 ms  
64 bytes from 10.77.4.20: icmp_seq=2 ttl=64 time=1.42 ms  
64 bytes from 10.77.4.20: icmp_seq=3 ttl=64 time=1.37 ms  
  
— 10.77.4.20 ping statistics —  
3 packets transmitted, 3 received, 0% packet loss, time 2003ms  
rtt min/avg/max/mdev = 1.374/4.280/10.050/4.080 ms  
  
=== No default internet route expected ===  
No IPv4 default route  
No IPv6 default route  
  
(kali@project04-kali)-[~]  
$
```

# Simulation Evidence

---

## Control objective

Create a limited, repeatable file-disruption scenario using disposable business-like data.

## What was done

A supplied lab account was used over SSH from Kali to Ubuntu. The script counted twenty synthetic invoices and checked for existing simulation markers. It renamed invoices 01 to 05 by appending .simulated-locked.

## What the evidence shows

The output reports five renamed files and lists the marked filenames alongside invoices that retain their original names. The controlled change was limited to five of the twenty sample documents.

## How this applies to client work

A comparable exercise would use approved test data in an agreed environment, with explicit limits, a tested recovery method and stop conditions. An active incident would trigger response procedures; additional simulation would not be introduced into the affected production service.

## What this evidence does not establish

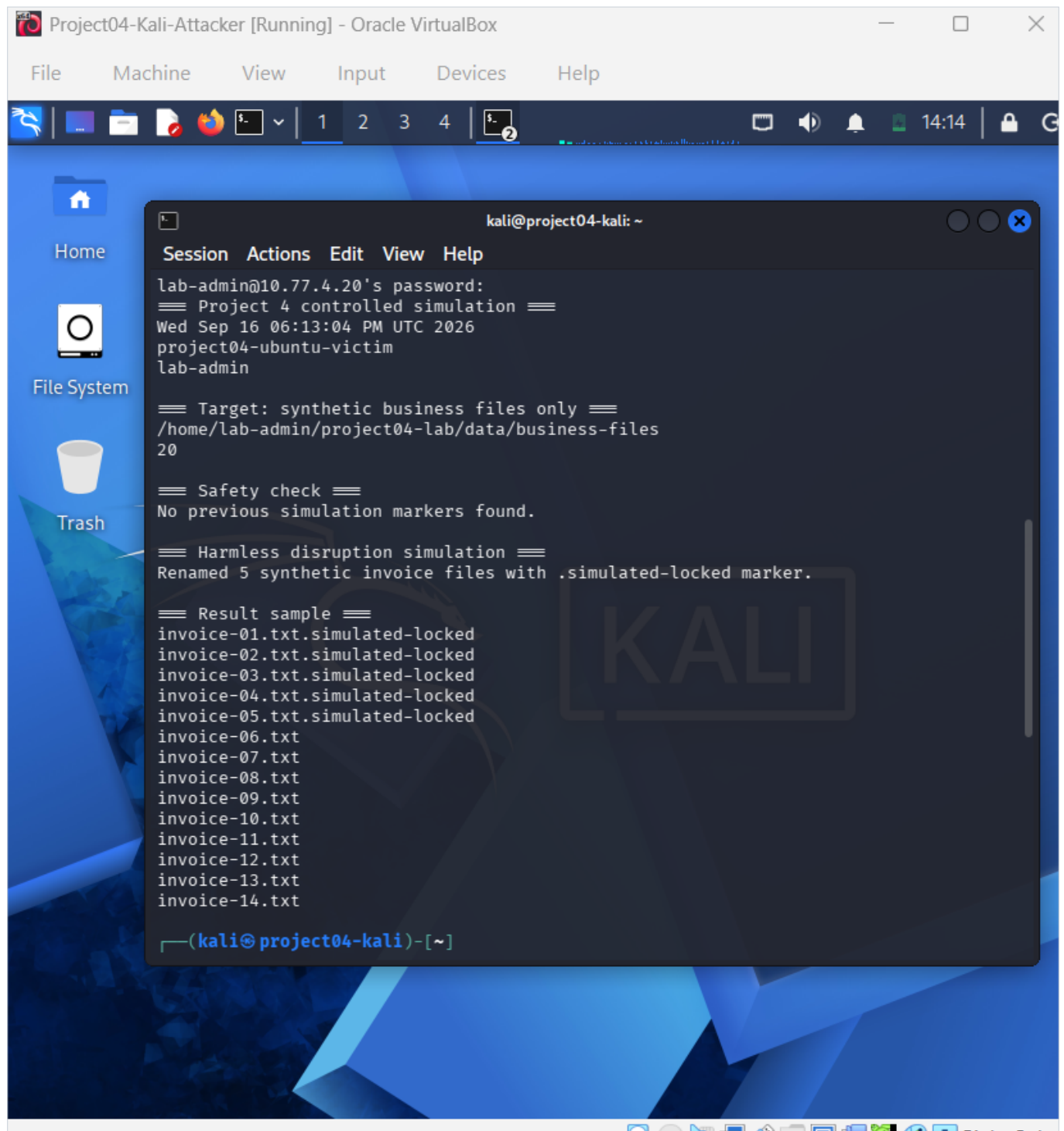
No real encryption, destructive malware, initial-access exploit or credential theft was performed. The marker changed filenames; it did not cryptographically lock the file contents. Business interruption was simulated, not measured against a live application.

**Supporting evidence: 1 image on the following page.**

## Simulation Evidence

### Five-file disruption simulation

The output records the safety check and five harmless rename markers on synthetic invoices.



```
kali@project04-kali: ~  
Session Actions Edit View Help  
lab-admin@10.77.4.20's password:  
=== Project 4 controlled simulation ===  
Wed Sep 16 06:13:04 PM UTC 2026  
project04-ubuntu-victim  
lab-admin  
  
=== Target: synthetic business files only ===  
/home/lab-admin/project04-lab/data/business-files  
20  
  
=== Safety check ===  
No previous simulation markers found.  
  
=== Harmless disruption simulation ===  
Renamed 5 synthetic invoice files with .simulated-locked marker.  
  
=== Result sample ===  
invoice-01.txt.simulated-locked  
invoice-02.txt.simulated-locked  
invoice-03.txt.simulated-locked  
invoice-04.txt.simulated-locked  
invoice-05.txt.simulated-locked  
invoice-06.txt  
invoice-07.txt  
invoice-08.txt  
invoice-09.txt  
invoice-10.txt  
invoice-11.txt  
invoice-12.txt  
invoice-13.txt  
invoice-14.txt  
  
(kali@project04-kali)-[~]
```

# Incident Timeline

---

## Control objective

Record observable checkpoints between the simulation and the final verified recovery.

## What was done

The Ubuntu identity and the five marked files were checked after the simulation. The displayed UTC times were compared with the simulation and recovery captures.

## What the evidence shows

The screenshots show simulation output at 18:13:04 UTC, an affected-file check at 18:31:26 UTC, and successful recovery verification at 20:28:51 UTC on 16 September 2026. These are recorded command-output checkpoints.

## How this applies to client work

Maintain a case log with source, timestamp, analyst, decision and approval for each major action. Verify clock alignment before calculating detection or recovery metrics. Record service impact and business recovery criteria separately from technical command results.

## What this evidence does not establish

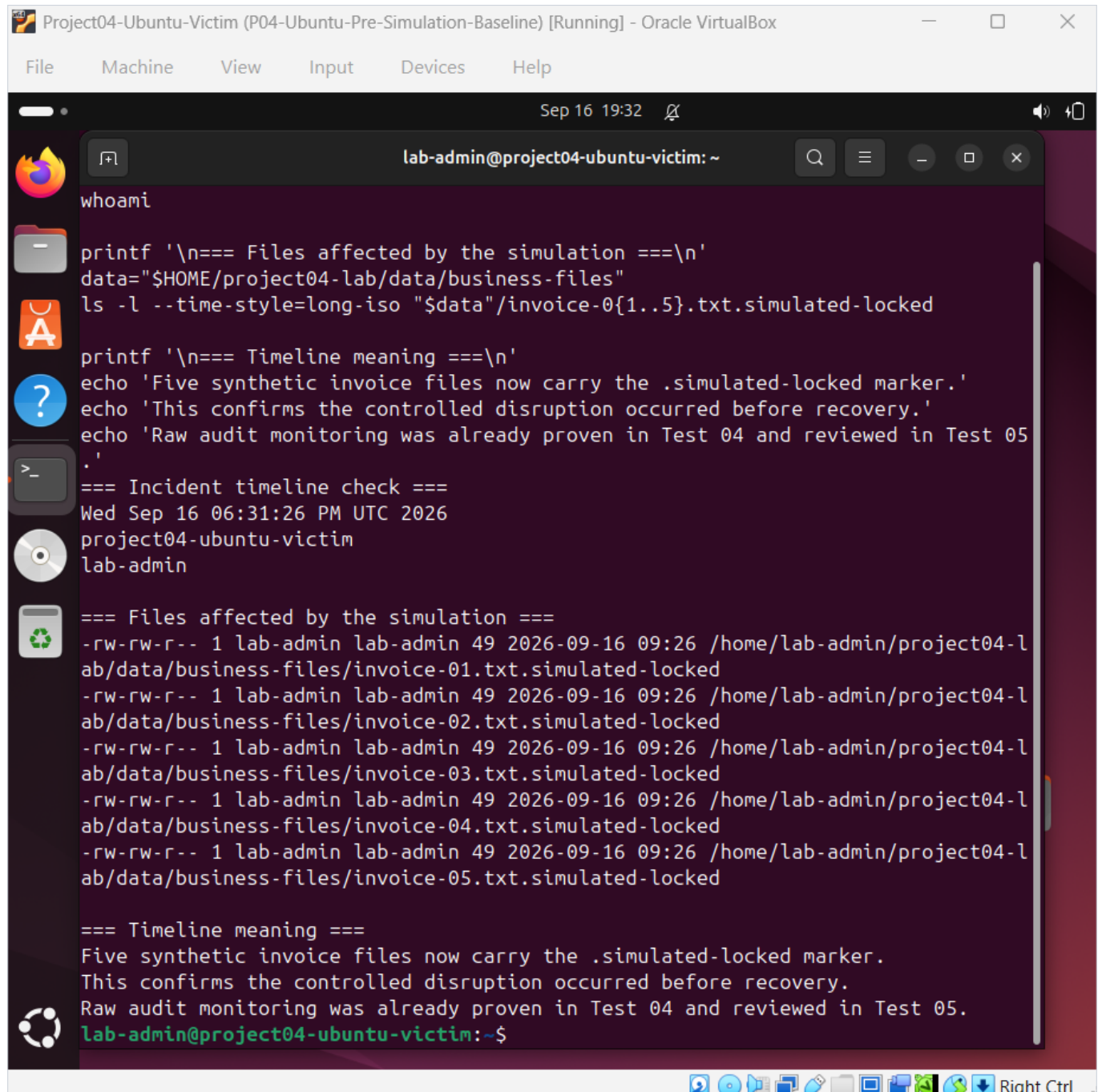
The file modification times shown in the directory listing are not proof of when a rename occurred. The explanatory echo lines are narrative, not independent audit events. Clock alignment and active work duration were not verified, so no response-time or recovery-time achievement is claimed.

**Supporting evidence: 1 image on the following page.**

## Incident Timeline

### Post-simulation affected-file checkpoint

The five marked files are present before recovery. The screenshot provides a state check, not a complete incident audit trail.



```
Project04-Ubuntu-Victim (P04-Ubuntu-Pre-Simulation-Baseline) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 16 19:32
lab-admin@project04-ubuntu-victim: ~
whoami
printf '\n=== Files affected by the simulation ===\n'
data="$HOME/project04-lab/data/business-files"
ls -l --time-style=long-iso "$data"/invoice-0{1..5}.txt.simulated-locked
printf '\n=== Timeline meaning ===\n'
echo 'Five synthetic invoice files now carry the .simulated-locked marker.'
echo 'This confirms the controlled disruption occurred before recovery.'
echo 'Raw audit monitoring was already proven in Test 04 and reviewed in Test 05'
.
=== Incident timeline check ===
Wed Sep 16 06:31:26 PM UTC 2026
project04-ubuntu-victim
lab-admin

=== Files affected by the simulation ===
-rw-rw-r-- 1 lab-admin lab-admin 49 2026-09-16 09:26 /home/lab-admin/project04-l
ab/data/business-files/invoice-01.txt.simulated-locked
-rw-rw-r-- 1 lab-admin lab-admin 49 2026-09-16 09:26 /home/lab-admin/project04-l
ab/data/business-files/invoice-02.txt.simulated-locked
-rw-rw-r-- 1 lab-admin lab-admin 49 2026-09-16 09:26 /home/lab-admin/project04-l
ab/data/business-files/invoice-03.txt.simulated-locked
-rw-rw-r-- 1 lab-admin lab-admin 49 2026-09-16 09:26 /home/lab-admin/project04-l
ab/data/business-files/invoice-04.txt.simulated-locked
-rw-rw-r-- 1 lab-admin lab-admin 49 2026-09-16 09:26 /home/lab-admin/project04-l
ab/data/business-files/invoice-05.txt.simulated-locked

=== Timeline meaning ===
Five synthetic invoice files now carry the .simulated-locked marker.
This confirms the controlled disruption occurred before recovery.
Raw audit monitoring was already proven in Test 04 and reviewed in Test 05.
lab-admin@project04-ubuntu-victim:~$
```

# Containment and Recovery

---

## Control objective

Preserve the marked copies, restore the missing originals and verify the complete invoice set.

## What was done

The marked files were moved out of the working invoice folder. The first restoration failed because its backup filename was empty; a misleading completion line was rejected. The retry used the recorded archive, verified its checksum, checked the preserved copies, restored the five originals and stopped on errors.

## What the evidence shows

The accepted screenshot shows the archive check passing, five marked copies preserved and all twenty invoices passing checks against the backup manifest. The restored file contents match that recorded baseline.

## How this applies to client work

Preserve necessary evidence, confirm the threat is contained, select a suitable clean recovery point and restore into a controlled environment. Validate data and business functions before the service owner approves return to use. See reference R3.

## What this evidence does not establish

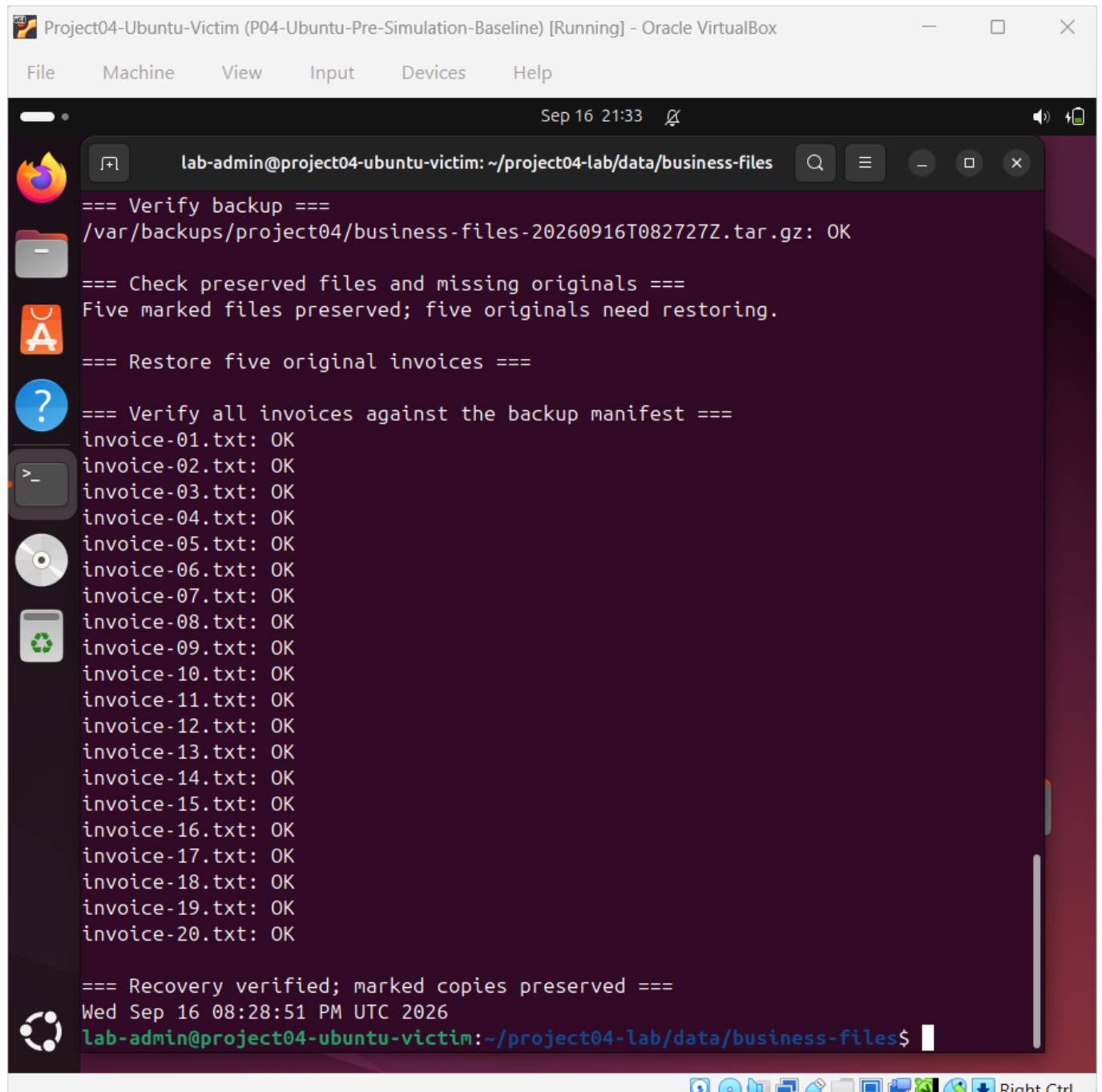
Moving harmless files aside is evidence preservation and working-folder cleanup. It does not prove isolation of a malicious process or host. The exercise verifies file recovery, not malware eradication, complete system recovery or production-service acceptance.

**Supporting evidence: 1 image on the following page.**

## Containment and Recovery

### Verified recovery of the invoice set

The corrected recovery passes the archive check and all twenty file checks while preserving the five marked copies.



The screenshot shows a terminal window titled "lab-admin@project04-ubuntu-victim: ~/project04-lab/data/business-files". The terminal output displays the following steps:

```
=== Verify backup ===  
/var/backups/project04/business-files-20260916T082727Z.tar.gz: OK  
  
=== Check preserved files and missing originals ===  
Five marked files preserved; five originals need restoring.  
  
=== Restore five original invoices ===  
  
=== Verify all invoices against the backup manifest ===  
invoice-01.txt: OK  
invoice-02.txt: OK  
invoice-03.txt: OK  
invoice-04.txt: OK  
invoice-05.txt: OK  
invoice-06.txt: OK  
invoice-07.txt: OK  
invoice-08.txt: OK  
invoice-09.txt: OK  
invoice-10.txt: OK  
invoice-11.txt: OK  
invoice-12.txt: OK  
invoice-13.txt: OK  
invoice-14.txt: OK  
invoice-15.txt: OK  
invoice-16.txt: OK  
invoice-17.txt: OK  
invoice-18.txt: OK  
invoice-19.txt: OK  
invoice-20.txt: OK  
  
=== Recovery verified; marked copies preserved ===  
Wed Sep 16 08:28:51 PM UTC 2026  
lab-admin@project04-ubuntu-victim:~/project04-lab/data/business-files$
```

The terminal window is part of an Oracle VM VirtualBox environment, as indicated by the title bar "Project04-Ubuntu-Victim (P04-Ubuntu-Pre-Simulation-Baseline) [Running] - Oracle VirtualBox". The window includes a menu bar (File, Machine, View, Input, Devices, Help) and a status bar at the bottom showing system icons and the text "Riaht Ctrl".

# Final Report

## Overall outcome

The lab demonstrated a controlled five-file disruption followed by verified restoration. All twenty sample invoices matched the recorded backup baseline after recovery. The five marked copies were preserved separately.

| Question                            | Evidence-backed answer                                                                                                                            |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Was there a usable recovery source? | Yes. The archive checksum passed and the recovered invoice set passed all twenty file checks.                                                     |
| Was monitoring demonstrated?        | Yes, for the separate controlled Test 04 file event. An alert for the Test 07 rename sequence was not demonstrated.                               |
| Was containment demonstrated?       | The lab network boundary was checked and marked copies were preserved. Active malicious-host or process isolation was not exercised.              |
| What business result is supported?  | The five original filenames were restored and the sample data matched its baseline. No production application or service-level target was tested. |

## Tools used and their role

**VirtualBox:** lab machines, internal networking and baseline snapshot. **Kali and OpenSSH:** controlled access using the lab account. **Ubuntu, auditd and ausearch:** file-event collection and review. **tar:** archive backup and extraction. **sha256sum:** archive and file-integrity checks.

## Recovery lesson

The initial restore had no usable archive filename and failed. Its printed completion message was not accepted as success. The corrected retry checked the archive first, stopped on errors and required successful file verification before reporting recovery.

# Authorized Client Workflow

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The following is a proposed delivery model for a real engagement. The client must approve the scope, access and response authority before work begins; the lab screenshots are not client authorization.

## 1. Agree authority and the working boundary

Identify the asset owner and approving representative. Record systems, environments, permitted methods, dates, exclusions, contacts and stop conditions. Agree who can approve disruptive actions, production changes and recovery. Testing permission does not automatically authorize every incident-response action. [R1]

## 2. Prepare recovery and evidence handling

The service owner identifies critical services and an acceptable recovery point. The response team checks the backup method, access and a test restore. Agree evidence storage, retention, client communications and any permitted AI use before accessing restricted material.

## 3. Validate monitoring and investigate

The analyst checks that an agreed test event reaches the intended review process. During an incident, correlate endpoint, identity, application and network records. Establish affected assets, business impact and confidence; preserve useful evidence and document decisions. [R2]

## 4. Contain under the response plan

The incident lead coordinates authorized endpoint or network isolation with the service owner, accounting for business dependencies. Preserve volatile evidence where feasible and verify containment. Moving files into another directory alone does not stop an active attacker. [R3]

### Required decision

The authorized client representative owns the decision to accept disruption, approve recovery and return a service to use. The analyst supplies the evidence and recommendation.

# Recovery and Handover

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## 5. Address the cause before reconnection

The response team investigates the entry path and persistence, applies approved corrective actions and documents any unresolved risk. A clean file checksum alone does not establish that the host is free of compromise.

## 6. Restore a suitable recovery point

Choose a backup from an appropriate point before the incident and assess its trustworthiness. Restore in a controlled environment and prioritise critical dependencies. For a compromised host, a rebuild may be needed before data is returned. Protected backups and controlled restoration are central to the CISA response guidance. [R3]

## 7. Verify data and business functions

Check file integrity, access permissions, application behaviour and representative business transactions. Record any missing or changed data and compare the outcome with the client's agreed recovery requirements. The service owner validates that operations can resume.

## 8. Close access and the evidence lifecycle

Revoke temporary accounts, sessions and access keys as applicable; rotate exposed credentials. A password change alone may not invalidate existing sessions. Return or retain evidence according to the agreement, record approved deletion and provide the final findings and retest status. [R4]

### Client deliverable

A concise case summary should state what happened, what was affected, what was restored, which checks passed, remaining risks, accountable owners and the approved next actions. Recommendations remain separate from completed work.

# Evidence Limits and Next Priorities

| Area              | Current evidence                                          | Next validation                                                                             |
|-------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Detection         | One benign file event was logged and reviewed.            | Capture the actual rename events, validate a detection rule and prove analyst notification. |
| Containment       | Internal lab networking and preservation of marked files. | Exercise and verify authorized host/process isolation and review residual access.           |
| Backup resilience | A checked archive and an off-VM copy on the same PC.      | Test a suitably independent recovery copy and the client's retention and access controls.   |
| Business recovery | Twenty sample invoice checks passed.                      | Test application behaviour, dependencies and service-owner acceptance.                      |
| Timings           | Selected command outputs display UTC checkpoints.         | Verify clock alignment and record detection, response and recovery milestones consistently. |

## Recorded sequence

On 16 September 2026, the simulation output displays **18:13:04 UTC**, the post-simulation file check displays **18:31:26 UTC**, and final verification displays **20:28:51 UTC**. The screenshots establish these checkpoints; they do not establish uninterrupted work duration or independently synchronised clocks.

## What can be claimed

The five-file simulation and checked file recovery are complete. No content mismatch was detected in the twenty invoices against the backup manifest. No claim is made of stopping real ransomware, demonstrating zero business data loss, meeting a recovery-time target or eliminating every security risk.

# Privacy and AI Use

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## Control objective

Explain the agency's proposed approach to client information when AI supports security work.

## What was done

A branded recommendations page was prepared using the confirmed company name, Anyaugo Space-Tech Ltd., and the existing logo. It covers workspace approval, data sharing, access, review, provider terms and accountable records.

## What the evidence shows

The guidance gives clients a practical basis for discussing how their data would be handled. It is a recommendation for adoption, not a claim that a client has already approved or implemented the controls.

## How this applies to client work

Agree the AI service, purpose, permitted data, access, retention and response contacts before use. Keep restricted information in the approved environment. Review sanitised inputs and AI-assisted outputs, and keep a record of approved disclosures. References R5 and R6 support the risk-management approach.

## What this evidence does not establish

A paid workspace, a local application or a written policy alone does not demonstrate privacy protection. Any supporting privacy app must be tested within the workflows it controls. No completed privacy-enforcement application is claimed in this project.

**Supporting evidence: 1 image on the following page.**

Six practical recommendations connect responsible AI use with data handling and human accountability.



**Anyaugo Space-Tech Ltd.**

SECURITY CONSULTING | CLIENT GUIDANCE

PROJECT 4 / TEST 11 / AGENCY RECOMMENDATIONS

# Client Privacy and Responsible AI Use

Our recommendation: keep client information inside an approved environment. Use AI for agreed purposes, with minimal disclosure, controlled access and human review.

|    |                                   |                                                                                                                                                                                                                                                                                                            |
|----|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | <b>Agree the workspace</b>        | Use the client's approved AI environment, or a company-managed workspace the client approves. Record the service, purpose and permitted data before use. Permission to test a system is separate from permission to share its information with an AI provider.                                             |
| 02 | <b>Limit what is shared</b>       | Use synthetic examples or reviewed, sanitised extracts. Keep passwords, keys and session tokens out of AI inputs. Keep customer records, private code and raw logs in approved storage unless a suitable processing arrangement is specifically approved. Check screenshots and file metadata too.         |
| 03 | <b>Control access</b>             | Use named accounts, multi-factor authentication and only the access needed for the engagement. Separate client work from personal work and other clients. Restrict connected folders, browsers and integrations; keep original evidence in access-controlled, encrypted storage.                           |
| 04 | <b>Review and verify</b>          | Have a responsible analyst review AI-assisted findings and recommendations against evidence. Approve production changes through the client's process, with backups and a rollback plan. Retest fixes and record the outcome before declaring an issue resolved.                                            |
| 05 | <b>Check provider terms</b>       | Confirm retention, training use, storage location and third-party processing before sharing restricted material. Agree return and deletion arrangements. An approved workspace still needs a privacy assessment; its name or subscription alone does not establish the required protections.               |
| 06 | <b>Keep an accountable record</b> | Record approvals, access, permitted disclosures, redactions and retest results. If unintended disclosure occurs, stop further sharing, preserve relevant records and notify the agreed client contact. At closure, remove access and document evidence return or deletion under the agreed retention plan. |

Project 4 uses synthetic invoices in a controlled lab. These are agency recommendations for client review and adoption; this exercise does not verify their implementation in a client environment.

References: NIST AI 600-1 (2024); NCSC Shadow IT guidance.

# References and Evidence Notes

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The references below inform the proposed client workflow. They are guidance, not permission to test an asset or certification of this lab. Links were checked or retrieved on 16 September 2026.

## **R1. NIST SP 800-115**

Technical Guide to Information Security Testing and Assessment (2008). Scope, rules of engagement, assessor skills and verification of remediation.

## **R2. NIST SP 800-61 Rev. 3**

Incident Response Recommendations and Considerations for Cybersecurity Risk Management (April 2025). Planning and integration of detection, response and recovery.

## **R3. CISA #StopRansomware Guide**

Preparation and response guidance, including coordinated isolation, evidence preservation and restoration from protected backups.

## **R4. OWASP Forgot Password Cheat Sheet**

Password-reset guidance includes invalidating existing sessions as a distinct action.

## **R5. NIST AI 600-1**

Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (2024). Considerations for AI risks, including privacy and human oversight.

## **R6. NCSC Shadow IT guidance**

Guidance on unmanaged tools and services, including unapproved AI used with organisational information.

## **Evidence provenance**

The fifteen embedded images come from the client evidence portfolio. Test 01 is a prepared scope record, Test 05 is a prepared analyst review and Test 11 is agency guidance; the other twelve are lab screenshots. Source images were preserved without retouching. The first failed recovery was observed in the troubleshooting screenshot supplied during the walkthrough; the retained Test 09 image shows the successful retry.

The report describes only the reviewed lab evidence and labels future client practice as proposed. No client credentials, production customer records or active malicious code were used in this exercise.